

# SQL Window Functions – Running Total (Interview Questions & Answers)

## 1. What is a Running Total?

**Answer:**

A **Running Total** (also called a cumulative total) is the progressive sum of values calculated row by row. Each row displays the sum of the current value and all previous values based on a specified order.

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## 2. Which SQL function is used to calculate a Running Total?

**Answer:**

The `SUM()` window function is used with the `OVER()` clause.

```
SUM(column_name) OVER(ORDER BY column_name)
```

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## 3. Why do we use `ORDER BY` inside the `OVER()` clause?

**Answer:**

`ORDER BY` defines the sequence in which SQL accumulates the values. Without an order, SQL cannot determine which rows come before or after, making a running total impossible.

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## 4. What is the difference between `SUM()` and `SUM() OVER()`?

**Answer:**

| <code>SUM()</code>                               | <code>SUM() OVER()</code>      |
|--------------------------------------------------|--------------------------------|
| Returns one aggregated result.                   | Returns a value for every row. |
| Reduces the number of rows.                      | Preserves all rows.            |
| Used with <code>GROUP BY</code> or entire table. | Used as a window function.     |

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## 5. What is the difference between `SUM() OVER()` and `SUM() OVER(ORDER BY ...)`?

Answer:

- `SUM() OVER()` returns the overall total on every row.
  - `SUM() OVER(ORDER BY ...)` returns a running (cumulative) total.
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## 6. Write a SQL query to calculate the running total of sales.

```
SELECT
    SaleDate,
    SalesAmount,
    SUM(SalesAmount) OVER(ORDER BY SaleDate) AS RunningTotal
FROM Sales;
```

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## 7. Can Running Total be calculated separately for each department?

Answer:

Yes. Use `PARTITION BY` to restart the running total for each department.

```
SELECT
    Department,
    EmployeeName,
    Salary,
    SUM(Salary) OVER(
        PARTITION BY Department
        ORDER BY EmployeeID
    ) AS RunningSalary
FROM Employees;
```

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## 8. What are some real-world uses of Running Totals?

Answer:

- Daily sales analysis
- Bank account balance tracking
- Inventory stock movement
- Monthly revenue growth
- Customer cumulative spending

- Cricket player's cumulative career runs
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## 9. Does Running Total remove duplicate rows?

**Answer:**

No. Running Total performs cumulative calculations while preserving every row.

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## 10. Interview Tip

**Question:** What is the difference between an aggregate function and a window function?

**Answer:**

An aggregate function combines multiple rows into a single result.

A window function performs calculations across a set of rows while keeping every row in the output.

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## Key Takeaways

- Running Total = Cumulative Sum
- Uses `SUM() OVER(ORDER BY ...)`
- `ORDER BY` determines the accumulation sequence.
- `SUM()` returns one result.
- `SUM() OVER()` repeats the overall total for every row.
- `SUM() OVER(ORDER BY ...)` returns a running total.
- `PARTITION BY` restarts the running total for each group.